Grade 7 Accelerated
Unit 2
Lesson 6 Practice

Name__Answer Key________
Date____________________
Period__________________

1. Identify an irrational number between -4 and -5.
   Answers will vary.

   Possible answers include:

   $-\sqrt{25} < x < -\sqrt{16}$, $\sqrt[3]{-125} < x < \sqrt[3]{-64}$

2. Victor thinks that $-\frac{28}{6}$ is greater than $\sqrt[3]{-100}$. Sean thinks that $\sqrt[3]{-100}$ is greater. Who is correct? Explain.

   $\sqrt[3]{-100} \approx -4.64$

   $\frac{-28}{6} = -4.\overline{6}$

   $\frac{17}{4} = 4.25$

   Sean is correct, $\frac{17}{4}$ is positive while $\frac{-28}{6}$ and $\sqrt[3]{-100}$ are both negative.

Compare each pair of expressions using <, >, or =.

3. $-2\pi + 10$ [○ < / ✓ > / ○ =] $\sqrt[3]{-56}$

4. $\sqrt{300}$ [✓ < / ○ > / ○ =] $\frac{196}{11}$

5. $-\sqrt{\frac{162}{2}}$ [○ < / ✓ > / ○ =] $-9.\overline{09}$

6. $3\sqrt{100}$ [✓ < / ○ > / ○ =] π^3

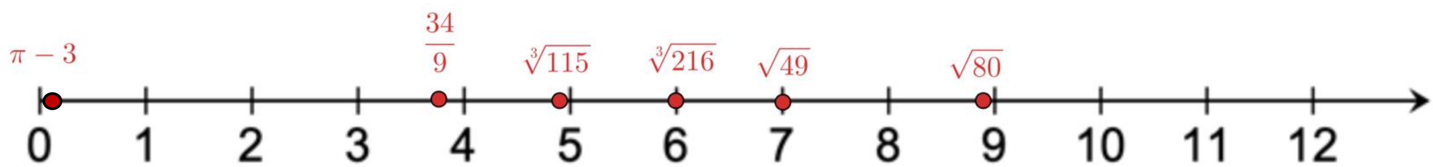
7. Order the following values from least to greatest.

$\pi - 3,$ $\sqrt{80},$ $\sqrt[3]{216},$ $\frac{34}{9},$ $\sqrt{49},$ $\sqrt[3]{115}$

≈ 0.1416 ≈ 8.94 $= 6$ $= 3.\overline{7}$ $= 7$ ≈ 4.86

$\boxed{\pi - 3, \frac{34}{9}, \sqrt[3]{115}, \sqrt[3]{216}, \sqrt{49}, \sqrt{80}}$

8. Plot and label each value from the previous question on the number line.



For questions 9-10, order each set of numbers from least to greatest.

9. $\sqrt{400},$ $\frac{78}{4},$ $2\pi^2$

$\frac{78}{4},$ $2\pi^2,$ $\sqrt{400}$

10. $\sqrt[3]{-126},$ $-\sqrt{25},$ $-5.0\overline{1}$

$\sqrt[3]{-126},$ $-5.0\overline{1},$ $-\sqrt{25}$